Ornob Barua

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Portfolio link: Ornob Barua | Mechanical Engineering Portfolio

EDUCATION

B.S./M.S. Mechanical Engineering, University of California, Santa Barbara B.S.: June 2027 M.S: June 2028
GPA: 3.84 Dean's Honors
Related Coursework: Mechatronics, Heat Transfer, Thermodynamics, Finite Element Analysis, Fluid Mechanics, Statics, Dynamical Systems, Machine Learning, MEMS fabrication, Vibrations, Heat Transfer, Design Optimization

EXPERIENCE

ASML **June 2026 - September 2026**
Mechanical Engineer Intern *San Diego, California*

- Developed angular arcmin tolerance stack up analysis for optical assembly to compare to requirement
- Established CMM workflow with MATLAB to conduct 3D tilt analysis to simulate predicted optical alignment.
- Created angular tilt calculator to enable theoretical tilt measurement based on GD&T/real life assembly
- Designed aluminum vee rail guide for CMM measurements, to reduce measurement uncertainty by ~5x

UCSB Nanofabrication Facility **March 2025 – March 2026**
Processing Technician Intern *Goleta, California*

- Operated scanning electron microscope (SEM) for imaging and analysis of different calibrations
- Performed FICP Si and SiO_2 calibrations using profilometry and ellipsometry to calculate selectivity and etch rate
- Executed photolithography using an ASML DUV stepper, followed by SEM analysis to verify pattern fidelity
- Performed ALD calibrations for Al_2O_3 and SiO_2 thin films, thickness verification with ellipsometry
- Completed calibrations for ICP-PECVD machine for SiO_2 and SiN thin films using metrology for analysis
- Developed maskless aligner (MLA) calibration process by redesigning calibration pattern with KLayout and Python, performing microscope and SEM-based analysis, and creating a traveler to guide new interns

Arista Networks **June 2025 – September 2025**
Mechanical Engineer Intern *San Jose, California*

- Designed and fabricated custom thermal test fixture for busbar power and return feeders
- Developed signal integrity test fixture for Time Domain Reflectometry measurements of line card and fabric module systems, enabling reliable debugging
- Utilized Creo Parametric and sheet metal design practices to model 20+ component for test fixture
- Created detailed part and assembly engineering drawings and issued drawings to vendors for quotation
- Tested Arista 7800 installation guide through full chassis assembly to improve clarity and usability

Cadense Group, UCSB **October 2024 – June 2025**
Undergraduate Research Assistant *Goleta, California*

- Developed synthetic shoe aging machine that has run for over 5 million cycles on 20+ shoes
- Engineered a rod rotation sensor system with electrical housing for motor power supply and Arduino Nano.
- Designed a custom motor bracket and wooden chassis with SolidWorks to ensure motor stability during operation

Microsoft **June 2024 - September 2024**
Garage Makerspace Intern *Mountain View, California*

- Provided technical support at the Microsoft Garage, assisting with prototyping and maintaining all equipment
- Co-facilitated and developed hands-on workshops integrating Arduino, 3D printing, and robotics, including projects such as Automatic Water Feeder, Line following robot, Silicon hand, Self-balancing robot, and RFID safe
- Produced concise and accurate prompts for Copilot to create and optimize Arduino code for all workshops

Projects

Exoskeleton Capstone Project | Team Lead **September 2026 – Present**
SiO_2/Glass/PDMS Microchannels **March 2026 – June 2026**

- Developed and followed process followers utilizing thermal oxidation, wet etching, contact photolithography, microscope, ellipsometer, and profilometer to fabricate 3 different types of microchannels
- Analyzed data from cleanroom metrology to fabricate chips to about ±0.2 micron of target dimensions
- Conducted current monitoring to test microchannels for microfluidic flow and compare to analytical results

SKILLS

Technical: Creo Parametric, SolidWorks, Fusion, GD&T, DFM, Cleanroom Manufacturing and Metrology, SEM, 3D printing, Arduino, MATLAB, COMSOL, Python, KLayout, Laser Cutting, Soldering, LaTeX, MS Excel
Professional: Presenting & Articulation, Teamwork, Time management, Adaptability, Problem Solving,